

Rift Chess

Move the piece.
Or move the ground beneath it.

14

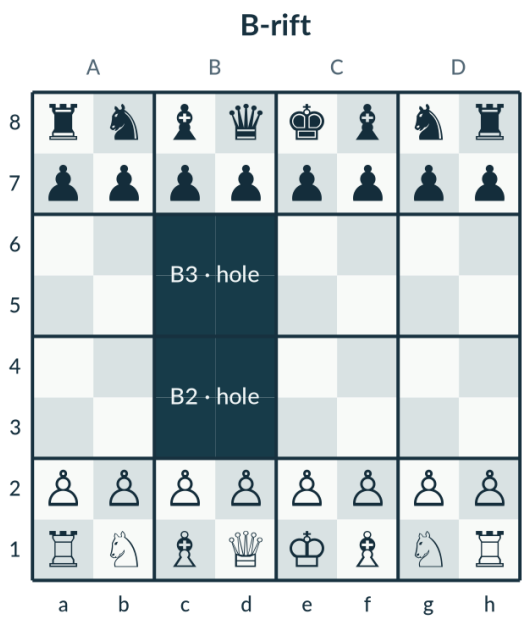
sliding 2x2 tiles

2

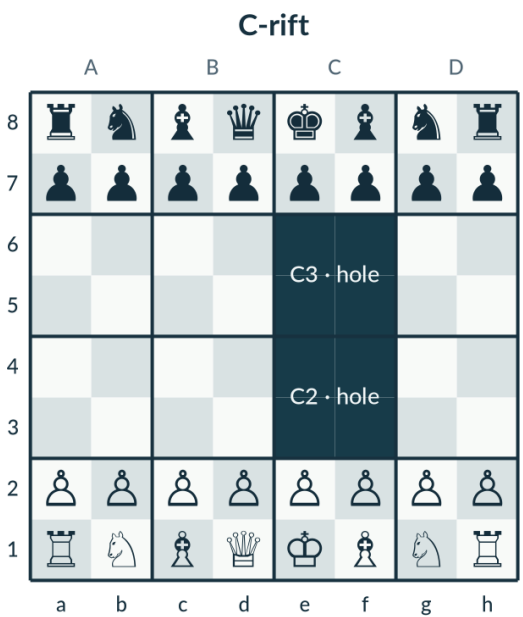
missing tile positions

1

action per turn



Missing c3:d6



Missing e3:f6

The decision: ordinary chess or one tile Shift per turn. Shift an empty tile or one friendly non-king passenger. Both kings anchor their current tiles.

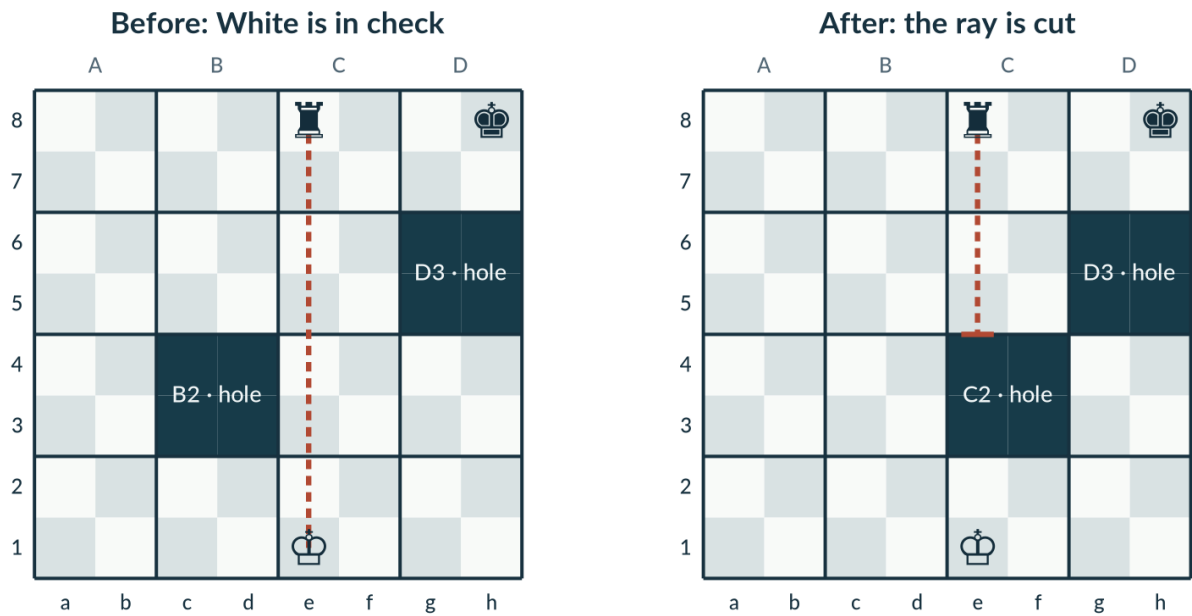
The central B/C openings each offer **15 ordinary moves + 4 Shifts**. They preserve rank-reflection/color-exchange geometry without removing starting pieces. Use both colors on the same layout for a rematch.

Ready for implementation, not a balance certificate. This package contains rules, an executable local engine, tests and replayable evidence. The intended client is a polished offline **3D** game; that client and its human experience are not yet built or measured.

Diagrams show exact rules geometry, not screenshots of a finished application.

The tactical idea

A Shift changes both position and the paths between pieces.



Shift C2 → B2. The empty tile moves; e3:f4 becomes a hole.

White is checked along the e-file. Moving the empty C2 tile into B2 removes e3:f4, so the rook can no longer reach the king. The passenger count is zero; the tactical effect is real.

Legal Shift = eligible load + adjacent hole + final king safety. A bright tile indicator must represent that complete test, not just a friendly passenger count.

Rays and gaps	Rooks, bishops and queens stop at a hole. Knights may jump intervening gaps, but their landing square must exist.
Transport	One passenger moves exactly two ranks/files. No capture occurs. A king never rides. Tile identities remain cosmetic.
Rights and promotion	Riding consumes pawn/rook movement rights. No Shift-created en passant. Promotion offers Q/R/B/N, including on a Shift.
Atomic state	Only the completed action determines check. Animation frames are not additional chess positions.

Fixture: cut_check_ray in fixtures/conformance.json. The full rulebook covers castling paths, effective en passant, backward pawn rides and draw precedence.

Correct the evidence first

The inherited tables were useful, but their “draw rate” hid unfinished games.

23.1%

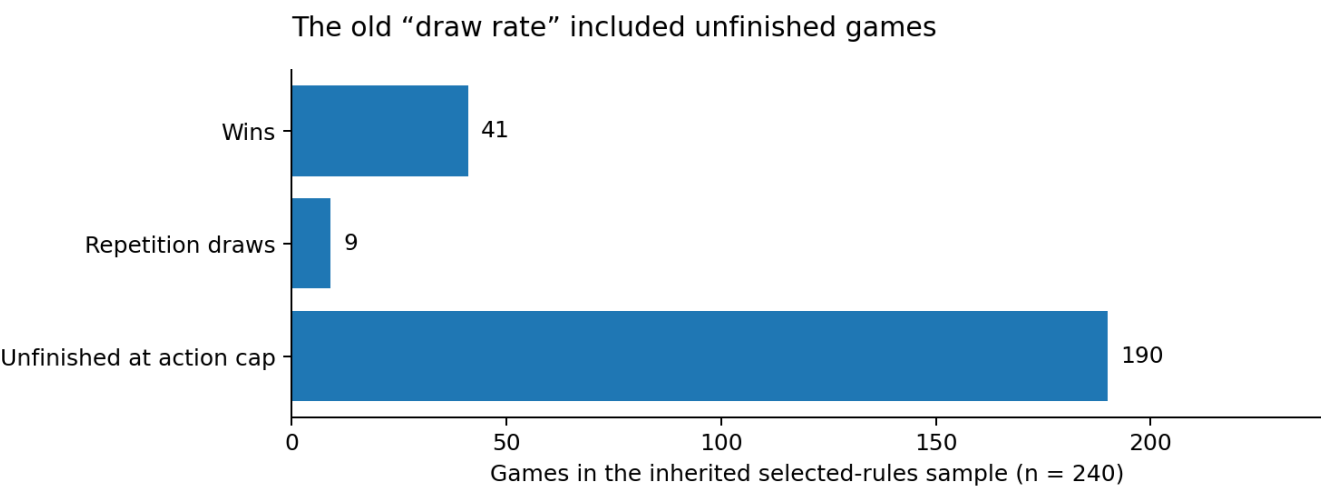
actions were Shifts

85.5%

of Shifts were empty

240

games in selected row



41 wins, 9 real draws, 190 unfinished. The old 82.9% “draw” rate included the simulation cap. It cannot be read as the rate at which real players would draw.

What the sample does suggest: empty tiles are doing much of the strategic work for these agents. The defining mechanic need not rely on moving multiple pieces in one turn.

The supplied research contains 19 aggregate rows describing **4,560 runs**. Their original per-game histories were not supplied. They remain clearly labeled as inherited, summary-only evidence.

Further audit notes: the unrestricted candidate changed several rules at once; two differently named layout rows share one geometry; first-Shift timing was zero-indexed. For the selected row, the player-facing average is approximately **action 14.8**.

Source: research/legacy/rules_summary.csv, exclusive_1. Neither Shift frequency nor a low repetition count is a direct measure of fun or strong-play balance.

Fresh tests, bounded claims

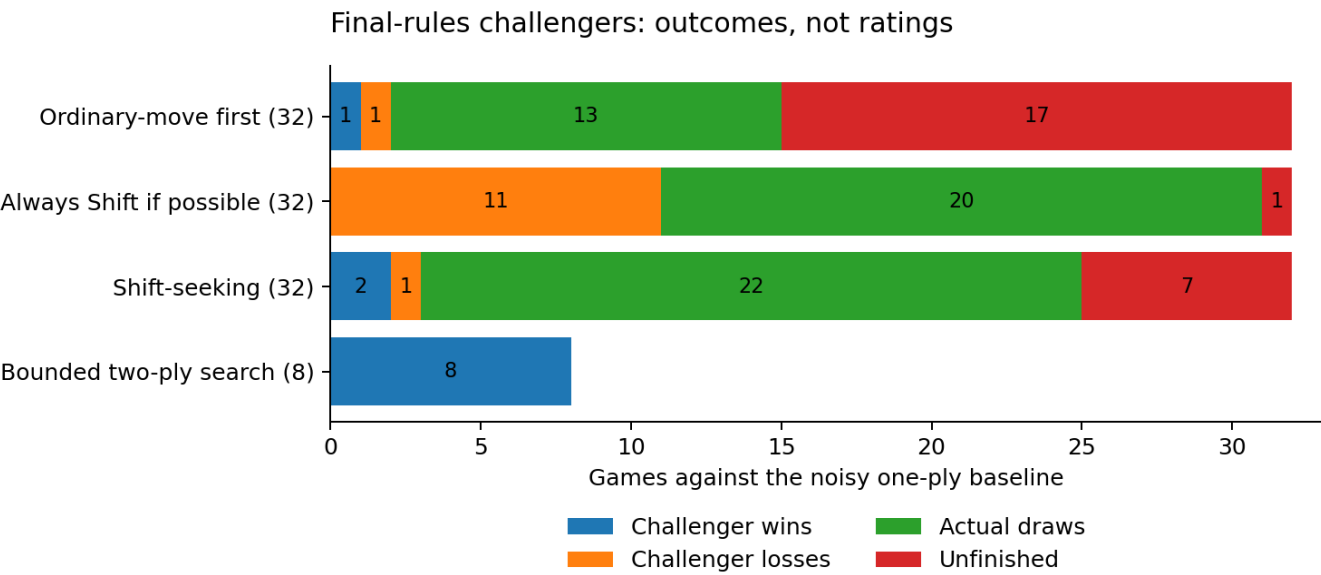
120 new final-rules games. Every history retained and replayed.

14,290

verified recorded actions

26 / 66 / 28

wins / draws / unfinished



Always Shift was exploitable: 0 wins, 11 losses, 20 repetition draws and 1 unfinished game against the baseline. There is an opportunity cost to ignoring ordinary moves in this test.

Ordinary-first was not shown to lose: 1–1 with 13 draws and 17 unfinished. Whether strong terrain play is essential remains unresolved. The Shift-seeking result is similarly inconclusive.

A real local bot is worthwhile: a bounded search targeting two plies won 8/8 against the weak one-ply baseline. This is a small engineering probe, not an Elo rating.

The separate 16-game baseline self-play sample used Shifts on 21.1% of actions: 2 wins, 11 draws, 3 unfinished. All groups used Prompted agreement; no agents offered draws. Shallow cap: 180 actions. Two-ply challenger cap: 100. Caps are never labeled draws.

Source: research/fresh/games.json and summary.json. Challenger colors and layouts were paired. Weak-policy results do not certify first-player or human balance.

Correctness and player choice

Rules and adjudication now have an executable local boundary.

67

passing tests

14 / 223

fixtures / successors

256

color-reflected positions

Check	Recorded result
Fresh replay	120 games / 14,290 actions
Random legal play	1,000 actions with invariant checks
Orthodox opening perft	20 / 400 / 8,902 / 197,281
API / opponent	Local NDJSON + bounded search
Action encoding	21,760 stable intent slots

At 100 quiet actions

Prompted agreement Default	Suggest a draw offer; continue unless both players agree. No unilateral claim.
Automatic at 100	End automatically, only when explicitly selected before the game.
No reminder	No progress-counter prompt or forced ending. Draw offers remain available.

All modes: threefold remains automatic; checkmate takes precedence over a same-action automatic draw. Captures, ordinary pawn moves and promotions reset the quiet counter; non-promoting pawn rides do not.

These are focused checks and same-engine replays. Fixtures establish parity, not independent proof. See `test_receipt.txt` and `verification.json` under `research/fresh/`.

The next test is the real game

Visual quality belongs in the launch gate, alongside rule correctness.

SCULPTURE & LIGHT

Three coherent environments. Two real piece geometry families. Convincing materials, tile thickness, contact and silhouette.

CAMERA & CLARITY

White, Black, Overview and Top, plus orbit. Move hints off; legal Shift cues on. Never confuse camera gestures with moves.

AUTHORED MOTION

Piece travel, rigid passenger Shifts, restrained captures and promotion. Reduced motion and responsive controls.

LOCAL & SHAREABLE

Packaged offline play and AI, local saves, headless API, GitHub source/releases, and a quiet confirmed donation link.

Still unknown: human learning cost, first-player advantage, strong-play exploits, endgame conversion, repeated-play appeal, 3D comfort and performance across hardware. No automated chart answers those by itself.

The build agent must first propose its stack/publication decisions for confirmation, then implement and conduct three genuine improvement passes: mechanics, visual inspection, and complete-game/offline distribution tests. A scaffold, a pretty image or unit tests alone do not satisfy completion.

The handoff is complete; the game is not yet built. The next deliverable should be the actual rendered application, played and improved, using this reference as a correctness tool.

Methods & provenance

Design: MDA (Hunicke et al.) and restricted play (Jaffe et al.). Ordinary rules: FIDE. Truncation: Farama. Origin: xkcd #3139.

Full sources, the clean decision narrative, raw fresh histories, commands, source hashes and tests are included. No cartoon artwork, third-party game code, human-study results or finished-client screenshots are represented as original project assets.